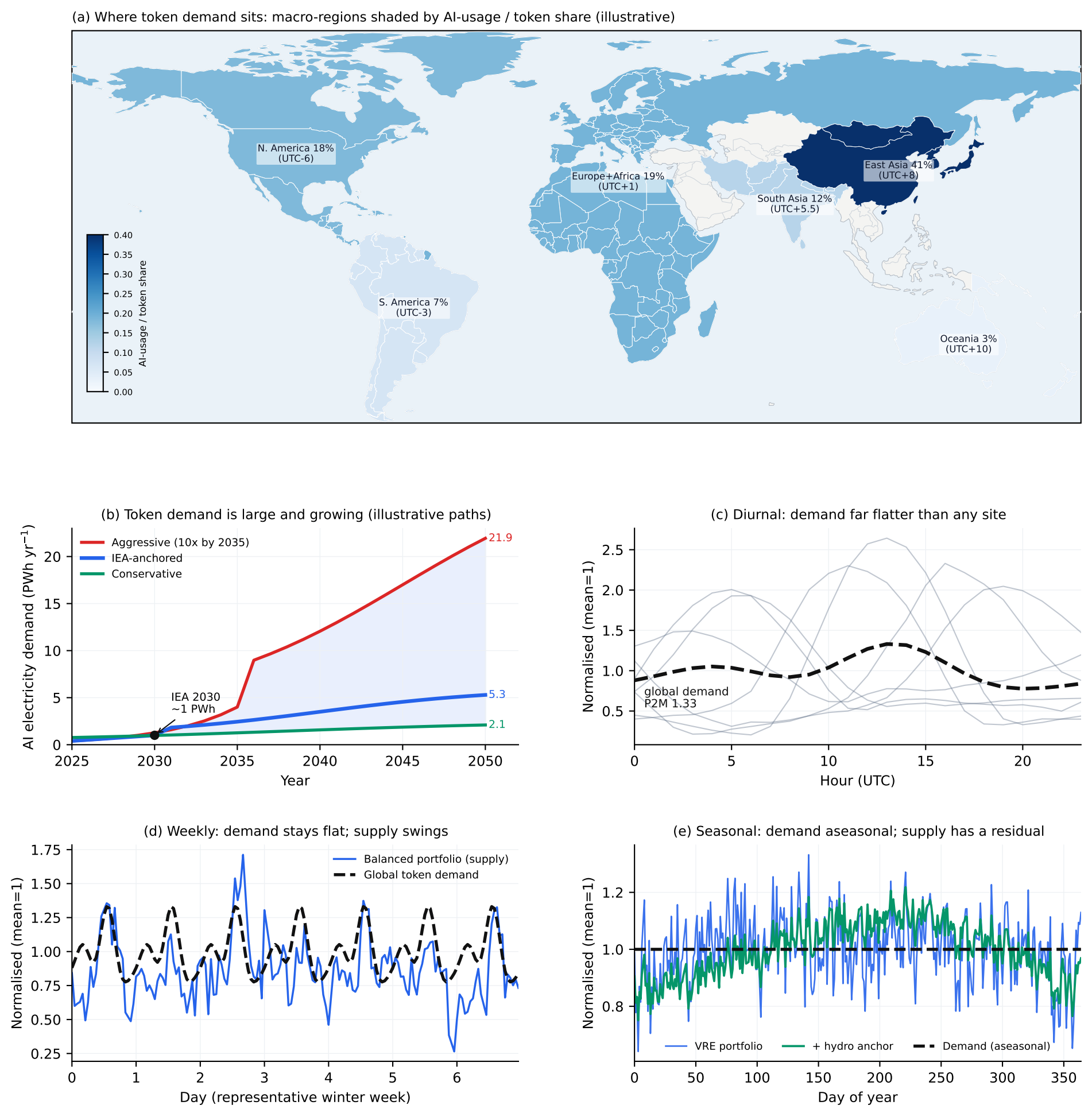

(a) Where token demand sits: macro-regions shaded by AI-usage / token share (illustrative)
N. America 18%
(UTC-6)
Europe+Africa 19%
(UTC+1)
East Asia 41%
(UTC+8)
South Asia 12%
(UTC+5.5)
S. America 7%
(UTC-3)
Oceania 3%
(UTC+10)
AI-usage / token share
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
(b) Token demand is large and growing (illustrative paths)
AI electricity demand (PWh yr⁻¹)
Aggressive (10x by 2035)
IEA-anchored
Conservative
21.9
5.3
2.1
IEA 2030
~1 PWh
Year
(c) Diurnal: demand far flatter than any site
Normalised (mean=1)
global demand
P2M 1.33
Hour (UTC)
(d) Weekly: demand stays flat; supply swings
Normalised (mean=1)
Balanced portfolio (supply)
Global token demand
Day (representative winter week)
(e) Seasonal: demand aseasonal; supply has a residual
Normalised (mean=1)
VRE portfolio
+ hydro anchor
Demand (aseasonal)
Day of year